

AP CSP | Lesson 24: Packets & Routing

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: B

B is the carried data. A is metadata; C is not the payload definition; D describes rules rather than message content.

Q2. Answer: C

C uses the specified sequence IDs. A uses arrival order; B reverses it; D ignores that text order matters.

Q3. Answer: A

A retains received pieces and reports the missing one. B claims success without all data; C discards valid pieces; D invents a replacement.

Q4. Answer: D

D states the limit. A adds ordered reliability; B invents fixed routing; C assigns guaranteed recovery to IP itself.

Q5. Answer: A and C

A describes TCP service; C identifies UDP limits. B mistakes reliability mechanisms for immunity; D contradicts dynamic routing. Select A and C only.

Q6. Answer: B

B is a task-dependent freshness tradeoff. A overgeneralizes; C promises performance without conditions; D denies possible recovery mechanisms.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: 1 "HEL"; 2 "LO "; 3 "AP "; 4 "CSP".

1 mark: Gives HEL for ID 1.

1 mark: Gives LO followed by one space for ID 2.

1 mark: Gives AP followed by one space for ID 3.

1 mark: Gives CSP for ID 4.

Q8. 4 marks

Answer: Present 4 and 2; missing 1 and 3; reconstruct 1,2,3,4; arrival order can differ from source order.

1 mark: Identifies IDs 4 and 2 as present.

1 mark: Identifies IDs 1 and 3 as absent.

1 mark: Gives order 1,2,3,4.

1 mark: Explains that sequence information restores source order despite variable arrival.

Q9. 4 marks

Answer: Incomplete; ID3 carries "AP "; retry the missing piece; verify all required IDs/payloads have arrived.

1 mark: States incomplete despite having some data.

1 mark: Identifies ID3 and its AP-plus-space payload.

1 mark: Describes resending/requesting the missing data.

1 mark: Requires receiver evidence of all required pieces before success.

Q10. 4 marks

Answer: S-A-D-T; 3 hops; no, T remains the destination; available connectivity does not guarantee packet delivery.

1 mark: Gives S-A-D-T.

1 mark: Counts three hops.

1 mark: Keeps destination T despite route change.

1 mark: Distinguishes route availability from guaranteed delivery.

Q11. 4 marks

Answer: TCP supports reliable ordered byte streams; acknowledgment/retransmission; UDP lacks delivery/order guarantees; persistent path or endpoint failure can prevent completion.

1 mark: States reliable ordered byte-stream service.

1 mark: Names acknowledgment or retransmission appropriately.

1 mark: Identifies a delivery or ordering guarantee UDP does not itself provide.

1 mark: Explains an ongoing failure can prevent successful completion.

Q12. 4 marks

Answer: An old position may be obsolete; exact file transfer; performance depends on conditions/application; model omits real acknowledgment, timeouts, congestion or failures of retries.

1 mark: Links a freshness priority to obsolete data.

1 mark: Gives a valid exact/ordered delivery task.

1 mark: Rejects universal speed claims with a relevant dependency.

1 mark: Names a specific omitted real protocol/network behavior.